

VERTEX CONNECTIVITY OF SOME CLASSES OF DIVISIBLE DESIGN GRAPHS

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Deza graphs were introduced in 1999 [1] as a generalisation of strongly regular graphs.

A k -regular graph on v vertices is called a **Deza graph** with parameters (v, k, b, a) , $b \geq a$ if the number of common neighbours of any two distinct vertices takes two values: a or b .

A Deza graph is called a **strictly Deza graph** if it has diameter 2 and is not strongly regular.

[1] M. Erickson, S. Fernando, W.H. Haemers, D. Hardy, J. Hemmeter, Deza graphs: A generalization of strongly regular graphs, *J. Comb. Des.*, **7** no. 6, (1999) 359–405

Divisible design graphs

A k -regular graph on v vertices is called a **divisible design graph** (DDG for short) with parameters $(v, k, \lambda_1, \lambda_2, m, n)$ if its vertex set can be partitioned into m classes of size n , such that two distinct vertices from the same class have exactly λ_1 common neighbors, and two vertices from different classes have exactly λ_2 common neighbors.

The definition implies that divisible design graphs are Deza graphs.

Feasible parameters of divisible design graphs

Divisible design graphs were first studied in master's thesis by M.A. Meulenberg [2].

In three following papers [3, 4, 5] DDGs were studied in more details and a lot of new constructions were presented.

[2] M.A. Meulenberg, Divisible Design Graphs, Master's thesis, *Tilburg University*, 2008.

[3] W.H. Haemers, H. Kharaghani, M.A. Meulenberg, Divisible Design Graphs, *J. of Comb. Th., Series A*, **118** (2011) 978–992

[4] D. Crnkovic, W.H. Haemer, More about Divisible Design Graphs, *CentER Discussion Paper*, **2011-140** (2011)

[5] D. Crnkovic, W.H. Haemer, Walk-regular divisible design graphs. *Designs, Codes and Cryptography*, **72** (2014) 165–175

Eigenvalues of divisible design graphs

Divisible Design Graphs graphs have at most 5 distinct eigenvalues.

Formulas for eigenvalues and their multiplicities are presented in the table:

Eigenvalues	Multiplicity	
$\theta_0 = k$	1	
$\theta_1 = \sqrt{k - \lambda_1}$	f_1	
$\theta_2 = -\sqrt{k - \lambda_1}$	f_2	$f_1 + f_2 = m(n - 1)$
$\theta_3 = \sqrt{k^2 - \lambda_2 v}$	g_1	
$\theta_4 = -\sqrt{k^2 - \lambda_2 v}$	g_2	$g_1 + g_2 = m - 1$

Equitable t -partition of a graph

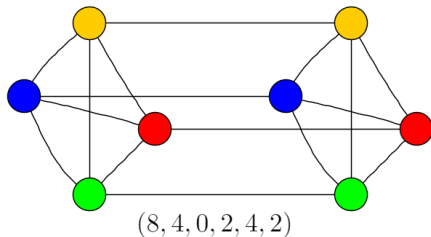
Let $V_1 \cup V_2 \cup \dots \cup V_t$ be the partition of the vertex set of a graph Γ with the property that every vertex of V_i has exactly r_{ij} neighbours in V_j .

Then $V_1 \cup V_2 \cup \dots \cup V_t$ will be an equitable t -partition of Γ .

Matrix $R = (r_{ij})_{t \times t}$ is called the quotient matrix of equitable partition.

Quotient matrix of a DDG

In case of divisible design graphs, the partition into m classes induce equitable m -partition. This partition is called **canonical** partition.



$$R = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

The **vertex connectivity** $\kappa(\Gamma)$ of the graph Γ is the minimum number of vertices whose deletion from Γ disconnects it.

For example, the vertex connectivity of a strongly regular graph is equal to its valency, which was proved by Brouwer and Mesner in 1985 ([6]). A similar statement for distance-regular graphs was proved by Brouwer and Koolen in 2009 ([7]).

[6] A. E. Brouwer and D. M. Mesner, The connectivity of strongly regular graphs, *Europ. J. Combin.* **6**, 215–216 (1985).

[7] A. E. Brouwer and J. H. Koolen, The vertex-connectivity of a distance-regular graph, *Europ. J. Combin.* **30** (3), 668–673 (2009).

Menger's theorem

Let x and y be two different vertices of a graph Γ . Two simple paths connecting x and y are called **disjoint** if they have no common vertices different from x and y .

A set of vertices S **disconnects** x and y if x and y belong to different connected components of the graph $\Gamma \setminus S$.

A set S of vertices of a graph Γ is called **disconnecting** if it disconnects some two of its vertices.

Menger's theorem

The minimum cardinality of a set disconnecting nonadjacent vertices x and y is equal to the largest number of disjoint paths connecting these vertices.

Dual Seidel switching

An involutive automorphism of a graph is called **Seidel automorphism** if it interchanges only non-adjacent vertices.

Permuting the rows (and not the columns) of the adjacency matrix of a graph according to Seidel automorphism is called **dual Seidel switching**.

Theorem 1 ([8, 9])

Let Δ be a Deza graph obtained from a strongly regular graph Γ with restricted eigenvalues $r > 0$ and $s < 0$ by dual Seidel switching. Then, we have one of the following two cases:

- (1) if $r \neq 2$, the vertex connectivity of Δ equals to its valency, except for the case when Γ is a complement of $n \times n$ -lattice, in that case the vertex connectivity is one less than valency;
- (2) $r = 2$.

[8] A.L. Gavriluk, S.V. Goryainov, V.V. Kabanov, On the vertex connectivity of Deza graphs, *Proc. Steklov Inst. Math.* 285 (Suppl 1) (2014) 68–77.

[9] S.V. Goryainov, D.I. Panasenko, On vertex connectivity of Deza graphs with parameters of the complements to Seidel graphs, *European Journal of Combinatorics* **80** (2019) 143–150.

The **lexicographic product** or **graph composition** $G[H]$ of graphs G and H is a graph such that the vertex set of $G[H]$ is $V(G) \times V(H)$ and adjacency defined by

$(u_1, u_2) \sim (v_1, v_2)$ if and only if $u_1 \sim v_1$ or $(u_1 = v_1 \text{ and } u_2 \sim v_2)$.

Construction 1.1

Let G be $K_{x \times y}$, the complete multipartite graph containing x parts of y vertices. Then $G[K_2]$ is a DDG with parameters $(2xy, 2y(x - 1) + 1, 2y(x - 1), 2y(x - 2) + 2, x, 2y)$.

DDGs with $\lambda_1 = k - 1$

Construction 1.2

Let G be a strongly regular graph with parameters (v, k, λ, μ) , where $\lambda = \mu - 1$. Then $G[K_2]$ is a DDG with parameters $(2v, 2k + 1, 2k, 2\mu, v, 2)$..

Construction 1.3

Let Γ be a DDG obtained with construction 1.2. Let M be the adjacency matrix of Γ , and P be a non-identity permutation matrix of the same size. Then PM is the adjacency matrix of a DDG if and only if P represents a Seidel automorphism.

These three constructions were considered in two papers [10, 11] on Deza graphs with parameters $(v, k, k - 1, a)$.

[10] V.V. Kabanov, N. Maslova, L. Shalaginov, On strictly Deza graphs with parameters $(n, k, k - 1, a)$, *European Journal of Combinatorics*, 80 (2019) 194–202.

[11] S. Goryainov, W.H. Haemers, V.V. Kabanov, L. Shalaginov, Deza graphs with parameters $(n, k, k - 1, a)$ and $\beta = 1$, *Journal of Combinatorial Designs*, 17(3) (2019) 188–202.

Vertex connectivity of DDGs with $\lambda_1 = k - 1$

Proposition 1

Vertex connectivity of a DDG with $\lambda_1 = k - 1$ equals $k - 1$.

Sketch of proof:

For a DDG Γ with $\lambda_1 = k - 1$ we have three cases:

(1) Γ obtained with Construction 1.1 and have parameters $(2xy, 2y(x - 1) + 1, 2y(x - 1), 2y(x - 2) + 2, x, 2y)$. In this case two non-adjacent vertices are (u_1, v_1) and (u_2, v_2) , where u_1 and u_2 are from the same part of $K_{x \times y}$ ($u_1 \neq u_2$) and $v_1, v_2 \in K_2$ (v_1 can be equal to v_2). Since u_1 and u_2 are from the same part of $K_{x \times y}$, (u_1, v_1) and (u_2, v_2) are from the same part of the canonical partition of Γ . Thus, (u_1, v_1) and (u_2, v_2) have $2y(x - 1)$ common neighbours, which gives $2y(x - 1)$ disjoint paths connecting these vertices. If we remove all $2y(x - 1)$ common neighbours, we have a disjoint union of edges, which means that by Menger's theorem the vertex connectivity of Γ equals to $2y(x - 1)$.

(2) Γ obtained with Construction 1.2 and have parameters $(2v, 2k + 1, 2k, 2\mu, v, 2)$. Note that Γ consists of two copies of G connected by several edges. Consider two vertices (u_1, v_1) and (u_1, v_2) , $v_1 \neq v_2$. They form a part of canonical partition of Γ . Thus, they have $2k$ common neighbours. If we remove all $2k$ common neighbours, we separate the edge formed by these vertices from the rest. So, $\kappa(\Gamma) \leq 2k$.

Considering all possible pairs on non-adjacent we can prove by Menger's theorem that $\kappa(\Gamma) = 2k$

(3) Γ obtained with Construction 1.3 and have parameters $(2v, 2k + 1, 2k, 2\mu, v, 2)$.

Γ can be viewed as follows (see [11]). Let Γ' be $G'[K_2]$, where G' is a Deza graph obtained from strongly regular graph G with parameters $(v, k, \lambda, \lambda + 1)$ by dual Seidel switching with Seidel automorphism ϕ . Modify Γ' as follows: for any transposition (xy) of ϕ , take the corresponding two pairs of vertices $x'x''$ and y', y'' in Γ' , delete the edges $\{x', x''\}$ and $\{y', y''\}$, and insert the edges $\{x', y''\}$ and $\{y', x''\}$. The resulting graph is exactly Γ .

Vertex connectivity of DDGs with $\lambda_1 = k - 1$

If G has an eigenvalues $r > 2$ and $s < -2$, then by Theorem 1 vertex connectivity of G' equals k and similarly to the case 2 vertex connectivity of Γ' equals $2k$.

In the case when $r \leq 2$ is noninteger, G has at most 25 vertices. The only strongly regular graphs satisfying this condition are the Paley graphs with parameters $(13, 6, 2, 3)$ and $(17, 8, 3, 4)$. By computer calculations these graphs don't have any Seidel automorphisms.

Vertex connectivity of DDGs with $\lambda_1 = k - 1$

In the case when $r = 1$ (also $s = -2$), G is one of Seidel graphs. There are three Seidel graphs with $\lambda = \mu - 1$: 3×3 -lattice with parameters $(9, 4, 1, 2)$, Petersen graph with parameters $(10, 3, 0, 1)$ and triangular graph $T(5)$ with parameters $(10, 6, 3, 4)$.

By computer calculations $T(5)$ doesn't have any Seidel automorphisms, one graph can be obtained by DSS from 3×3 -lattice and one graph can be obtained by DSS from Petersen graph. Vertex connectivity of DDGs obtained from these two graphs equals $2k$.

In the case when $r = 2$ (also $s = -3$) there exist 26 SRGs with $\lambda = \mu - 1$ (see [12]): 15 graphs with parameters $(25, 12, 5, 6)$, 10 graphs with parameters $(26, 10, 3, 4)$ and one graph with parameters $(50, 7, 0, 1)$ (Hoffman-Singleton graph). By computer calculations, vertex connectivity of DDGs obtained from these graphs equals $2k$.

Thus, all DDGs obtained by Construction 1.3 have vertex connectivity equal to $2k$.

[12] V.V. Kabanov, A.A. Makhnev, D.V. Paduchikh, On strongly regular graphs with eigenvalue 2 and their extensions, *Doklady Mathematics*, **81** (2010) 268–271.

Hadamard matrices

An $m \times m$ matrix H is a **Hadamard matrix** if every entry is 1 or -1 and $HH^T = mI$. A Hadamard matrix H is called **graphical** if H is symmetric with constant diagonal, and **regular** if all row and column sums are equal.

Construction 2 ([3, Construction 4.8])

Consider a regular graphical Hadamard matrix H of order $l^2 \geq 4$ with diagonal entries -1 and row sum l . The graph with adjacency matrix

$$A = \begin{bmatrix} M & N & O \\ N & O & M \\ O & M & N \end{bmatrix},$$

where

$$M = \frac{1}{2} \begin{bmatrix} J + H & J + H \\ J + H & J + H \end{bmatrix} \text{ and } N = \frac{1}{2} \begin{bmatrix} J + H & J - H \\ J - H & J + H \end{bmatrix},$$

is a DDG with parameters $(6l^2, 2l^2 + l, l^2 + l, (l^2 + l)/2, 3, 2l^2)$.

Proposition 2

Vertex connectivity of a DDG obtained with Construction 2 is no more than $2l^2$.

Sketch of proof:

Consider notations from Construction 2.

Say x is the number of 1s in each row of H and y is the number of -1 s in each row H , that gives $x - y = l$ and $x + y = l^2$.

Each row of $\frac{1}{2}(J + H)$ has x 1s and each row of $\frac{1}{2}(J - H)$ has y 1s. Thus each row of A has $2x + x + y$ 1s, which equals $2l^2 + l$.

Vertex connectivity of DDGs

Consider $A' = \begin{bmatrix} M & O \\ O & N \end{bmatrix}$, which can be obtained by removing $2l^2$ rows from A .

A' gives a disconnected graph with two components: $(l^2 + l)$ -regular graph and l^2 -regular graph.

Thus, vertex connectivity of DDGs obtained with Construction 2 is no more than $2l^2$.

- (1) Find the exact value of the vertex connectivity of DDGs from Construction 2.
- (2) Find the vertex connectivity of other families of DDGs.
- (3) Find some bounds for the vertex connectivity of DDGs in general.

Thanks for your attention!